

WEEK 10: ORGAN SYSTEM

INTEGUMENTARY SYSTEM

- Largest organ externally

- **Largest internally:** liver
- Comprises 15-20% of total body weight
- **Total surface thickness:** 1.5 to 2 m²
- **Average thickness:** 1 to 4 mm
 - Thickest: palms, soles
 - Thinnest: eyelids, lips, genital

- Composed of skin and *its accessory structures*

- Hairs, nails, sweat glands and sebaceous glands

- First line of defense from external environment

FUNCTIONS

- 1.) **Protection** - barrier against microorganisms, UV radiation, and mechanical injury
- 2.) **Insulation** - internal heat and thermal barrier
- 3.) **Receptors** - sensory receptors for touch, pressure, pain, temperature, and vibration
- 4.) **Aids in homeostasis** - internal balance
- 5.) **Temperature regulation** - via sweat evaporation and vasodilation/vasoconstriction
- 6.) **Elimination of waste products** - through perspiration (urea, salts, water)
- 7.) **Synthesis of vitamin D** - converts 7-dehydrocholesterol to vitamin D3 in sunlight
- 8.) **Oil production** - through sebaceous glands
- 9.) **Fluid retention** - prevents excessive water loss and dehydration
- 10.) **Cosmesis (appearance)** - physical identity and social perception

MAIN COMPONENTS

1.) Skin (cutaneous membrane)

- Epidermis: outer epithelial layer
 - Keratin, melanin
- Dermis: inner connective tissue layer
 - Loose areolar connective and adipose tissue
- Hypodermis (subcutaneous layer)
 - Not part of the skin proper, but connects it to the underlying tissue

2.) Accessory (appendage) structures:

- Hair and hair follicle
- Nails
- Glands:
 - Sebaceous (oil) glands
 - Sweat glands - eccrine (thermoregulation) and apocrine (scent)

LAYERS OF THE SKIN

EPIDERMIS

- Outermost protective layer
- *Avascular* - nourishment from dermis
- Composed of keratinized stratified squamous epithelium
- Thickness:
 - Thick skin: palms and soles (≈ 0.4-1.55 mm; contains stratum lucidum)
 - Thin skin: eyelids, face, body (≈ 0.07-0.12 mm)
 - Contains more pain receptors

MAJOR CELL TYPES OF THE EPIDERMIS

1.) Keratinocytes — “main structural cells”

- Located in stratum spinosum
- Produce keratin, a tough, fibrous protein for protection
- Form ~90% of the epidermal cells
- Undergo continuous keratinization (formation -> migration -> desquamation)

2.) Melanocytes — “pigment-producing cells”

- Located in stratum basale
- Produces melanin for skin color and UV protection

- Darker = less chances of carcinoma

3.) Langerhans cells — “immune defense cells”

- Located in stratum spinosum from bone marrow
- Act as antigen-presenting cells (warning system) to trigger immune responses

4.) Merkel cells — “sensory receptors”

- Located in stratum basale (especially in fingertips and lips)
- Mechanoreceptors for light touch sensation

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NUMBER OF CELL LAYERS OF EPITHELIAL TISSUE

1.) Simple epithelium

- Single cell layer
- Function: absorption, secretion, and diffusion
- Location: intestines, capillaries, alveoli,

2.) Stratified epithelium

- Two or more layers
- Function: protection against friction and wear
- Location: skin, esophagus, mouth

3.) Pseudostratified epithelium

- Appears multilayered due to nuclei at varying levels, but all cells touch the basement membrane
- Location: trachea

4.) Transitional epithelium

- Multilayers that can stretch or change shape
- Function: specialized for distension
- Location: urinary bladder, ureter

CELL SHAPE OF EPITHELIAL TISSUE

1.) Squamous

- Shape: flat, thin cells
- Function: Diffusion and filtration
- Common in alveoli and endothelium

2.) Cuboidal

- Shape: cube-shaped cells with central nuclei
- Function: Secretion and absorption
- Common in kidney tubules, glands, and ducts

3.) Columnar

- Shape: tall, column-like cells
- Function: Secretion and absorption
- Common in intestinal lining

Epithelial Type	Description	Location / Example	Function
Simple Squamous	Single flat layer	Alveoli, capillary walls, lining of blood vessels (endothelium)	Diffusion, filtration
Simple Cuboidal	Single cube-shaped layer	Kidney tubules, thyroid follicles	Secretion, absorption
Simple Columnar	Single tall layer, may have microvilli or goblet cells	GI tract, gallbladder	Absorption, secretion
Pseudostratified Columnar (Ciliated)	Appears layered, but all cells touch basement membrane	Trachea, upper respiratory tract	Mucus secretion, ciliary movement
Stratified Squamous (Non-keratinized)	Multiple layers, moist surface	Oral cavity, esophagus, vagina	Protection from abrasion
Stratified Squamous (Keratinized)	Multiple layers with keratinized surface cells	Epidermis of skin	Protection, water resistance
Stratified Cuboidal	Two layers of cube-shaped cells	Sweat glands, mammary glands	Secretion, protection
Stratified Columnar	Rare, several layers of columnar cells	Male urethra, salivary ducts	Secretion, protection
Transitional Epithelium	Stratified; cells change shape during stretching	Urinary bladder, ureters	Allows distension and recoil

CLINICAL RELEVANCE

1.) Metaplasia

- Reversible change of one epithelial type to another
- E.g., pseudostratified to squamous in smokers
- E.g., Barret's - non-keratinized to simple squamous

2.) Dysplasia

- Abnormal cell growth; may progress to neoplasia (cancer)
 - Irreversible, pre-cancerous change, disorganized
 - Autonomous cell proliferation

3.) Carcinoma

- Malignant tumor arising from epithelial tissue
- E.g., basal cell and squamous cell carcinoma

LAYERS OF THE EPIDERMIS — “CuLuGo Sa Batok”

1.) Stratum corneum

- Outermost layer
- Composed of flattened, dead keratinocytes (corneocytes) filled with keratin
- Where callus happens
 - From destroyed eleidin
- Function: waterproofing and protection (prevents water loss)

2.) Stratum lucidum

- Clear, thin layer of dead, densely packed corneocytes filled with eleidin
- Found in palms and soles

3.) Stratum granulosum

- 3-5 layers of flattened keratinocytes containing keratohyalin (for keratin) and lamellar bodies (waterproof barrier)
- Marks the beginning of keratinization and cell death

4.) Stratum spinosum

- 8-10 layers of keratinocytes connected by desmosomes, giving a “spiny” appearance
- Contains Langerhans cells
- Function: Strength and flexibility

5.) Stratum basale

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- Single layer of cuboidal/columnar basal cells attached to the basement membrane
- Function: site of mitosis, melanocytes, and Merkel cells

DERMIS — corium or “true skin”

- Beneath epidermis and forms the bulk of the skin's thickness
 - 20-30 times thicker than the epidermis
- Composed of connective tissue for strength, elasticity, and extensibility
- Provides structural support, vascular supply, and nutrients to epidermis
- Serves as a site for sensory nerve endings, blood and lymphatic vessels, and appendages structures

MAIN COMPONENTS

1.) *Fibers*

- Collagen (≈70%) - provides tensile strength and resistance to mechanical stress
- Elastin - allows elastic recoil and flexibility
- Reticular fibers - form a supporting network

2.) *Ground substance*

- Mucopolysaccharides (glycosaminoglycans) and proteoglycans
 - Retain water, maintain turgor (ability to stretch and go back to original shape), and hydration

3.) *Cells*

- Fibroblasts - produce collagen and elastin (baby cells)
- Macrophages & mast cells - immune defense and inflammation control
- Adipocytes - fat storage (mainly in deeper dermis and hypodermis)

LAYERS OF THE DERMIS

1.) *Papillary layer — superficial*

- Loose (areolar) connective tissue
- Features: capillary loops, Meissner's corpuscles (touch), free nerve endings
- Function: nutrition to epidermis, for sensation and thermoregulation

2.) *Reticular layer — deep*

- Dense irregular connective tissue rich in collagen & elastin fibers
- Features: sweat glands, sebaceous glands, hair follicles, Pacinian corpuscles (pressure)
- Functions: strength, elasticity, and anchorage for appendages

BLOOD SUPPLY OF THE DERMIS

1.) *Superficial capillary plexus*

- Location: papillary layer of the dermis
- Function: nutrients and oxygen to epidermis through simple diffusion

2.) *Deep vascular plexus*

- Location: reticular layer and junction of dermis and hypodermis
- Function: reservoir for blood volume regulation and supplies larger structures such as sweat and sebaceous glands, and hair follicles

NERVE SUPPLY OF THE DERMIS

1.) *Free nerve endings*

- Location: throughout dermis and epidermis
- Function: nociception, temperature, and itch

2.) *Meissner corpuscle*

- Location: papillary layer (fingertips, lips, eyelids)
- Function: light touch and texture discrimination

3.) *Pacinian corpuscle*

- Location: deep in the reticular layer and hypodermis
- Function: deep pressure and vibration

4.) *Ruffini ending (corpuscle)*

- Location: reticular dermis and joint capsules
- Function: sense skin stretch, sustained pressure, and grip modulation

5.) *Merkel disc*

- Location: stratum basale
- Function: fine touch and shape/edge discrimination

MOTOR INNERVATION

- Supplied by ANS (sympathetic fibers) that controls:

- Sweat gland secretion (sudomotor activity)
- Arrector pili muscle contraction (piloerection) - *smallest smooth muscle*

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- Stapedius - smallest skeletal muscle
- Vasomotor tone (blood vessel constriction/dilation)
 - Temperature and BP regulation

RETE PEGS — “fingerprints” & “footprints”

- Finger-like projections where the epidermis interdigitates with the dermis

- Functions:

- Increase surface area for nutrient exchange (through diffusion)
- Strengthen the junction between epidermis and dermis
- Help the skin resist shear forces and friction

THERMOREGULATION

1.) Vasodilation

- Increased blood flow to the skin -> heat loss through radiation and sweat evaporation

2.) Vasoconstriction

- Reduced blood flow -> heat conservation
- Controlled by ANS (sympathetic fibers)

CLINICAL RELEVANCE

1.) Ehlers-Danlos syndrome

- Connective tissue disorder

2.) Flattening of Rete Pegs

- In elderly, it decreases frictional resistance (skin tears more easily)

3.) Flushing or blushing

- Vasodilation in the superficial plexus due to emotion or temperature

- E.g., pallor rash, erythema, rosacea

3.) Pallor

- Vasoconstriction or reduced cutaneous blood flow

4.) Cyanosis

- Bluish discoloration due to hypoxia in the dermal capillaries

5.) Pressure ulcers

- Impaired perfusion in dermal vessels from prolonged compression
- Pain receptors may be destroyed

6.) Peripheral neuropathy

- Loss of cutaneous sensation due to nerve

damage

- E.g., diabetes mellitus

7.) Sensory testing in PT

- Evaluation of dermatomes and sensory receptor integrity during neurological assessment

8.) Burns

- May destroy dermal nerve endings -> loss of sensation

- High risk of sepsis

9.) Tactile acuity

- Areas with dense receptor populations (fingertips, lips) have higher sensory precision

SUBCUTANEOUS TISSUE — “hypodermis” or “superficial fascia”

- Deepest layer -> beneath dermis

- Composed of adipose tissue and loose (areolar) connective tissue

- Not part of the skin proper, but functionally and structurally connects to it

- Transition zone between the skin and the underlying muscles, tendons, and bones

MAIN COMPONENTS

1.) Adipose tissue

- Cushioning, insulation, and energy storage

2.) Loose connective tissue

- Contains collagen and elastin fibers for flexibility and mobility of the skin

3.) Large BV and lymphatics

- Major routes for nutrient exchange and temperature regulation

4.) Nerves

- Large branches of cutaneous nerves and sensory receptors

FUNCTIONS OF HYPODERMIS

1.) Support

- Anchors the skin to underlying structures to aid in movement without tearing

2.) Insulation

- Adipose tissue regulates body heat

3.) Cushion

- Provides shock absorption to prevent mechanical trauma

4.) Storage

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- Serves as an energy reserve for triglycerides that can be metabolized for energy during fasting

APPENDAGES

HAIR

LAYERS OF THE SKIN SHAFT

1.) *Cuticle*

- Outermost thin layer of overlapping keratinized cells

- Function: provides protection

2.) *Cortex*

- Middle layer containing keratin and melanin
- Function: for hair strength, elasticity, and color

3.) *Medulla*

- Central core; may be continuous, fragmented, or absent
- More prominent in coarse hair

TYPES OF HAIR

1.) *Terminal hair*

- Coarse, thick, pigmented
- Found on scalp, eyebrows, axilla, and pubic
- Due to influence (especially androgens)

2.) *Vellus hair*

- Fine, short, non-pigmented
- Covers most of the body (e.g., face, limbs, chest) -> *most common in females*
- Function: for thermoregulation and protection

FUNCTIONS OF HAIR

1.) *Protection*

- Shields scalp from UV rays and injury
- Eyelashes and nasal hairs protect eyes and nostrils from dust

2.) *Sensory function*

- Hair follicles contain nerve ending (root hair plexuses) sensitive to light touch

3.) *Thermal regulation*

- Prevents heat loss in cold environments

4.) *Aesthetic role*

- Appearance and social/sexual identity

ARRECTOR PILI MUSCLE

- Type: smooth muscle attached to the hair follicle

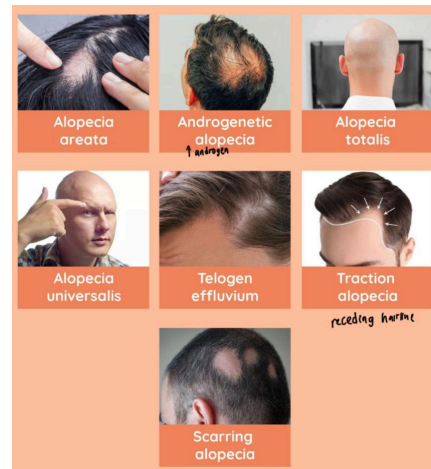
- Function: causes piloerection when stimulated by cold, fear, or emotional stimuli
- Mechanism: Contraction pulls the hair upright and compresses the sebaceous gland, aiding in sebum secretion

CLINICAL RELEVANCE

1.) *Baldness*

- From terminal to vellus hair
- Due to increased dihydrotestosterone (DHT)
- Treatment: finasteride to decrease testosterone

2.) *Alopecia*



NAILS

- Hard keratinized plates covering the dorsal surface of the distal phalanges of fingers and toes
- Modified epidermal appendages, derived from stratum basale and spinosum

STRUCTURES OF THE NAILS

1.) *Nail plate (body)*

- Visible, hard portion composed of compact keratinized cells
- Rests on the nail bed

2.) *Nail bed*

- Skin beneath the nail plate, vascular

3.) *Lateral nail folds*

- Skin folds that overlap the sides of the nail plate for protection and support

4.) *Proximal nail fold (cuticle/eponychium)*

- Fold of the skin at the base of the nail plate

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- Protects the matrix from infection and debris

5.) Lunula

- Crescent-shaped whitish area at the nail base
- Represents the visible portion of the nail matrix

6.) Nail root (matrix)

- Hidden portion under the proximal nail fold
- Site of nail growth and keratin production

7.) Hyponychium

- Area of thickened epidermis under the free edge of the nail
- Acts as a barrier to pathogens

FUNCTIONS OF THE NAILS

1.) Protection

- Shields from trauma

2.) Fine motor control

- Provides counterpressure, enhancing tactile sensitivity and precision grip

3.) Aesthetic/social function

- Reflects health and hygiene; often used for cosmetic purposes

4.) Clinical indicator

- Can indicate systemic disease
- E.g., cyanosis, clubbing, spoon nails

CLINICAL RELEVANCE

1.) Onychomycosis

- Fungal infection of the nail causing thickening and discoloration

2.) Koilonychia

- Spoon-shaped nails associated with iron-deficiency anemia

3.) Clubbing

- Nail angle > 180°
- Seen in chronic hypoxia (cyanosis)
- E.g., COPD, congenital heart disease
- Schamroth sign
 - Determinant of clubbing; diamond shape

4.) Beau's lines

- Transverse grooves due to temporary growth arrest (fever, malnutrition, trauma)

TYPES OF SWEAT GLANDS

1.) Eccrine (merocrine)

- Widely distributed (palms, soles, forehead)
- Function: thermoregulation via sweat evaporation

2.) Apocrine

- Found in axillary, genital, and areolar regions
- Function: body odor due to bacterial action from vitamin C, proteins, and lipids during emotional stress

SEBACEOUS (OIL) GLANDS

- Secretes sebum
 - Lubricates skin and hair
- Function: prevent dryness, waterproofing, antibacterial protection

CLINICAL RELEVANCE

- 1.) Whiteheads (open comedones) - blockage
- 2.) Blackheads (closed comedones) - oxidized
- 3.) Acne vulgaris - infected

SWEAT (SUDORIFEROUS) GLANDS

- Open at the skin pore

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URINARY SYSTEM

- Maintains homeostasis of body fluids by filtering the blood, removing waste products, and regulating fluid, electrolyte, and acid-base balance

- **Composed of:**

- **Kidneys**
 - Filter blood, removes wastes, and produce urine
- **Ureters**
 - Transport urine from kidneys to bladder
- **Urinary bladder**
 - Temporarily stores urine
- **Urethra**
 - Conducts urine to the exterior of the body

FUNCTIONS

- 1.) **Homeostasis** - maintains water, electrolyte, and acid-base balance
- 2.) **Osmoregulation** - Controls osmolarity of body fluids
- 3.) **Metabolic waste excretion** - Eliminates ammonia, creatinine, urea, uric acid
- 4.) **Endocrine function** - secretes erythropoietin (RBC production), renin (BP control), and calcitriol (vitamin D activation/calcium balance)

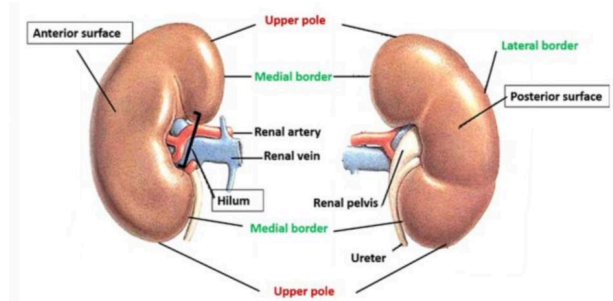
KIDNEYS

LOCATION & EXTERNAL ANATOMY

- Retroperitoneal on the posterior abdominal wall (T12-L3)

- Right kidney: slightly lower due to liver
- Shape: bean-shaped
- Size: 11 x 6 x 3 cm; ~150 g each
- **Surfaces:**
 - **Anterior:** convex and faces anterolaterally
 - **Posterior:** flat and faces posteromedially
- **Borders:**
 - **Medial:** concave, and has a vertical cleft in the middle (hilum)
 - **Lateral:** convex
- **Poles:**

- **Superior:** nearer the median plane than the inferior pole (*near diaphragm & suprarenal gland*)
- **Inferior:** more laterally placed as compared to the superior pole and is about 1 inch above the iliac crest (*near intestinal lobe and psoas major*)
- **Hilum:** entrance/exit for renal artery, renal vein, and ureter



PROTECTIVE COVERING

inside -> outside

1.) Renal capsule

- Tough, fibrous covering adherent to kidney surface; prevents trauma and infection

2.) Perirenal fat capsule

- Surrounds the capsule; cushions and insulates the kidney

3.) Renal fascia (Gerota's fascia)

- Anchors kidney to posterior abdominal wall

4.) Pararenal fat

- Outermost adipose tissue layer providing extra protection

- Kidney punch: intervention if kidney is inflamed

- If bile secretion is low, fat will go to kidney
- UTI may lead to pyelonephritis (inflammation of kidney due to pus)

INTERNAL ANATOMY OF KIDNEY

A. Two main regions

1.) Renal cortex

- Outer, granular layer containing renal corpuscle and convoluted tubules

2.) Renal medulla

- Inner region organized into renal pyramids

- Ends in renal papillae (apex of pyramid), that project medially to minor calyces

B. Collecting system

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1.) *Minor calyces*

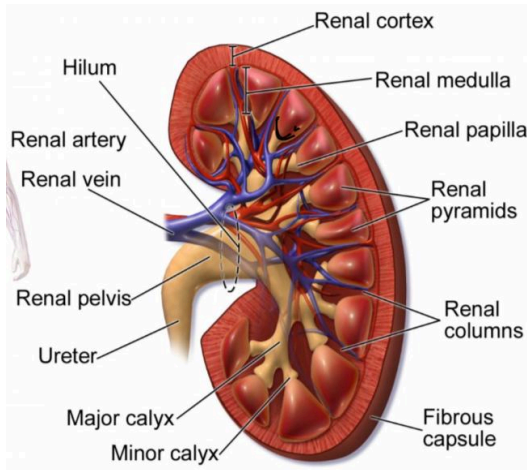
- Collect urine from papillae

2.) *Major calyces*

- form from merging of minor calyces

3.) *Renal pelvis*

- Collects urine from major calyces and continues as ureter



- Regulated by ADH (for water and Na reabsorption)
- Controls final urine concentration

TYPES OF NEPHRONS

1.) *Cortical nephron*

- Location: outer cortex
- Loop of Henle: short
- Function: major site of filtration; 85% of nephrons

2.) *Juxtamedullary nephron*

- Location: near medulla
- Loop of Henle: long
- Function: concentrates urine; 15% of nephrons; prevents water loss and dehydration

BLOOD SUPPLY

- Do not anastomose; do not connect
- Dilated BV = increased BP = hypotension
- Cardiac output/TBV received by the kidney is 25% (1.5 L)

1.) *Renal arteries*

- Arise from abdominal aorta between the level of L1 and L2

2.) *5 segmental branches (arteries)*

- Divides at the hilum at the kidney

3.) *Lobar arteries*

- Divides at the renal sinus

4.) *Interlobar arteries*

- Lobar arteries further divides
- Pass between pyramids in the medulla

5.) *Arcuate arteries*

- Divides at the cortico-medullary junction

6.) *Interlobular arteries*

- Pass through the cortex

7.) *Afferent arterioles*

- Interlobular arteries further divides which feed glomerular capillaries

8.) *Efferent arterioles*

- Ending of afferent arterioles

8.) *Peritubular capillary plexus*

- Around collecting tubules and ducts
- Eventually travels to PCT and DCT

NEPHRON — structural & functional unit

- Responsible for filtration, reabsorption, and secretion
- Each kidney contains 1-1.5 million nephrons

Components:

1.) *Renal corpuscle*

- Site of filtration; initial process of urine formation
- Composed of glomerulus (collection of blood) and Bowman's capsule (filtering of blood)

2.) *Proximal convoluted tubule (PCT)*

- Reabsorbs 65-70% of filtered water, Na, glucose, amino acids, and proteins

3.) *Loop of Henle*

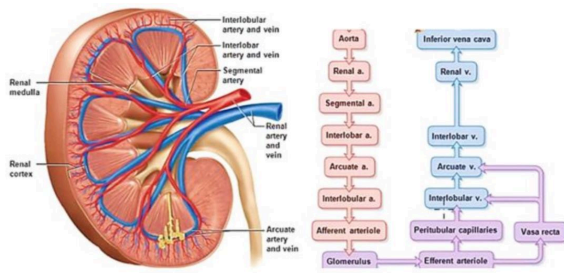
- Establishes osmotic gradient
 - Descending limb - water reabsorption; concentrated; if blocked, nephrolithiasis forms
 - Ascending limb - Na and Cl reabsorption; diluted

4.) *Distal convoluted tubule*

- Active secretion and fine-tuning of electrolytes (Na, K, Ca)

5.) *Collecting duct*

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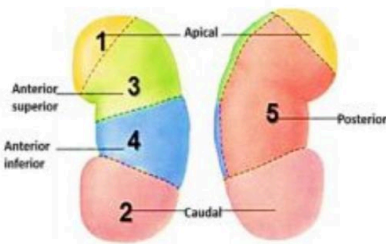
VENOUS DRAINAGE

- Peritubular capillary plexus drains into interlobular veins -> arcuate -> interlobar -> segmental -> renal veins -> IVC
- **Left renal vein** receives blood from left suprarenal and left gonadal veins

Vascular Segments of Kidney

Each kidney has five vascular segments

1. Apical/superior
2. Caudal/inferior
3. Upper anterior
4. Middle Anterior
5. Posterior



URETERS

- Pair of muscular tubes (~25-30 cm long)
- Transports urine from the renal pelvis to the urinary bladder
- **Retroperitoneal**
 - Descend anterior to the psoas major, cross the pelvic brim (medial to the renal pelvis), and enter the bladder posterolaterally at an oblique angle

WALL LAYERS

1.) Mucosa

- Transitional epithelium (urothelium)
 - Allows stretching as urine passes
- Lamina propria
 - Loose connective tissue beneath the epithelium

2.) Muscularis

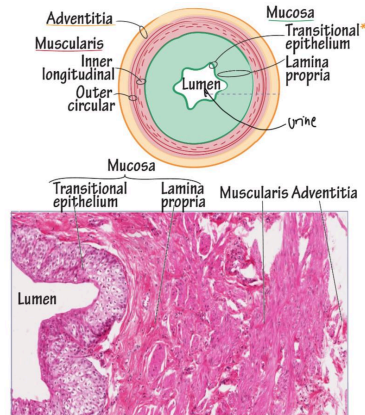
- Two layers of smooth muscle; reverse arrangement to the small intestine
 - Inner longitudinal

- Outer circular

- Lower (distal) third may have an additional outer longitudinal layer for stronger contractions

3.) Adventitia

- Connective tissue that anchors the ureter to surrounding structures
- Contains BV, lymphatics, and nerves supplying the ureter



FUNCTION OF THE URETER

- Transport urine from the kidneys to the bladder through peristaltic contractions every 10-15 seconds, which could be assisted by gravity and hydrostatic pressure
 - Ureterovesical junction (UVJ) - functional valve to prevent backflow (vesicoureteral reflux) during bladder contraction

SITES OF URETERAL CONSTRICTION

- Sites for renal/ureteral stone
- 1.) **Ureteropelvic junction**
 - Where renal pelvis narrows into ureter
- 2.) **Ureterovesical junction**
 - Where ureters enters the bladder wall obliquely
- 3.) **Pelvic brim**
 - Where ureter crosses over the common iliac artery

BLOOD SUPPLY & NERVE SUPPLY — both ureter and urinary bladder

1.) Arterial supply

- Upper third - renal arteries
- Middle third - gonadal (below SMA) and common iliac arteries
- Lower third - superior and inferior vesical arteries

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2.) Venous drainage

- Corresponds to arterial supply
- Internal iliac vein via vesical venous plexus

3.) Nerve supply

- Autonomic fibers from renal, aortic, and hypogastric plexuses
- Pain fibers follow sympathetic nerves (T11-L2) -> referred pain to flank, groin, or genital (UTI)
 - Parasympathetic (S2-S4) - contract detrusor muscle, relax urethral sphincter, micturition
 - Sympathetic - relax detrusor muscle, contract sphincter, prevent micturition

URINARY BLADDER

- Muscular, distensible sac that serves as a temporary reservoir for urine
- Lies in the pelvic cavity, posterior to the pubic symphysis
 - In males: located anterior to the rectum and superior to the prostate gland
 - In females: Located anterior to the vagina and inferior to the uterus
- **Average capacity:** 400-600 mL
 - **First urge:** 150-200 mL
 - **Strong urge:** 350-400 mL
 - Can stretch beyond 1 liter in pathologic distension
- **Shape:**
 - **Empty:** pyramidal shape (lies within pelvis)
 - **Full:** ovoid and rises into the abdominal cavity
- **Lining:** transitional epithelium - for stretching and protection
- **Muscle wall:** detrusor muscle - composed of three interlacing layers
 - **Inner and outer longitudinal, and middle circular**
 - Contracts during micturition to expel urine

INTERNAL FEATURES

1.) Trigone

- Smooth triangular area on the bladder floor

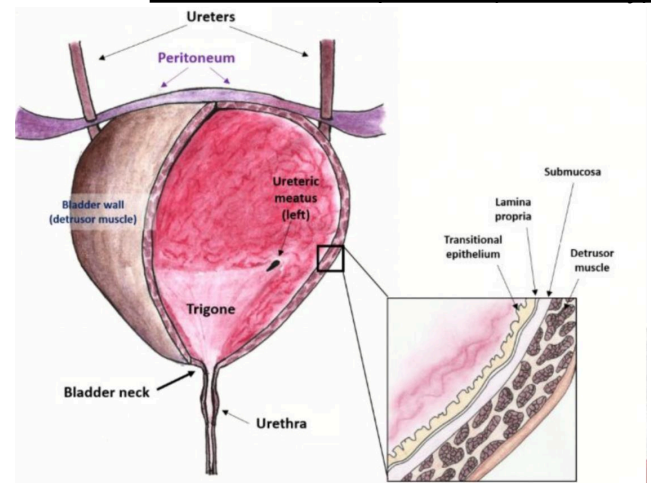
- Bounded by:

- Two ureteric opening (posterolateral corners)
- Internal urethral orifice (inferior apex)

- Function: serves as a funnel for urine outflow
- Remains smooth even when bladder is empty (non-distensible)

2.) Neck of the bladder

- Lowest part and most fixed part, surrounding the internal urethral orifice
- Contains internal urethral sphincter (involuntary)



FUNCTIONS OF URINARY BLADDER

- Temporarily stores urine and expels it

1.) Storage phase

- Detrusor muscle relaxes
- Internal and external sphincter contract to retain urine

2.) Voiding (micturition) phase

- Controlled by parasympathetic activation (S2-S4)
 - If full, causes dribbling
 - Contains Onufrius nucleus (Onuf's nucleus)
- Detrusor muscle contracts, sphincters relaxes
- Voluntary control via external urethral sphincters (pudendal nerve, S2-S4)

3.) Normal urination frequency: 4-6 times

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CLINICAL REVELANCE

1.) *Trigonitis*

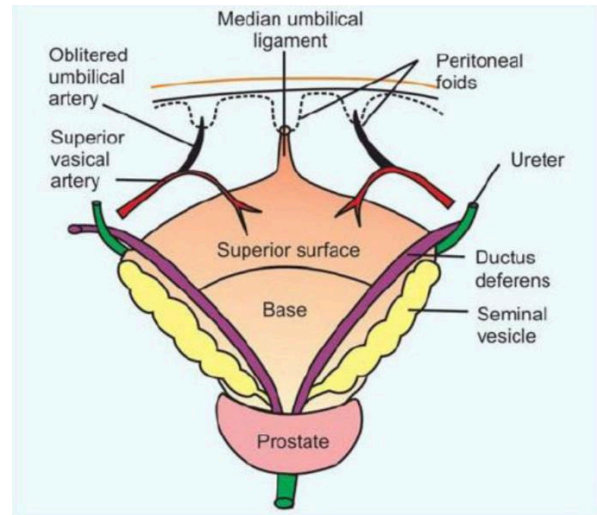
- Inflammation of trigone; most common site of infection
- From UTI bacteria

2.) *Urinary tract infection (UTI) — “ascending infection”*

- Most commonly caused by *E. coli* (starts with trigonitis)
- Painful urination, lower abdominal discomfort; may ascend to kidneys causing fever, flank pain (pyelonephritis)
- Can lead to kidney stones

3.) *Nephrolithiasis — “kidney stones”*

- Solid crystalline deposits forming in the kidney or ureter
 - Urine becomes supersaturated with minerals (calcium, oxalate, uric acid)
 - Mainly caused by dehydration, high dietary minerals, UTI, urease-producing bacteria
- Flank pain (renal colic), hematuria, obstruction, nausea/vomiting, severe bleeding rapture (peritonitis)



BLOOD & NERVE SUPPLY

1.) *Arterial supply*

- Superior and inferior vesical arteries (from internal iliac artery)
- Female: vaginal artery also contributes

2.) *Venous drainage*

- Internal iliac vein via vesical venous plexus

3.) *Lymphatic drainage*

- Internal and external iliac lymph nodes

4.) *Nerve supply*

- Sympathetic (T11-L2)
 - Causes bladder to relax and sphincter to contract
- Parasympathetic (S2-S4)
 - Cause bladder to contract and sphincter to relax for micturition
- Somatic (Pudendal nerve, S2-S4)
 - Controls external urethral sphincter